

Normalised remainders of a finite log-polynomial expansion

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Abstract

Deterministic half of Astra #11 rank 4 (the ordered-remainder statement of `thm:strataempiricalalexpansion`). Given $Z(N) = \sum_{\gamma \in P} N^{-\gamma} (A_\gamma \log N + B_\gamma) + R(N)$ with $|R(N)| \leq KN^{-E}(1 + \log N)$, bounded coefficients $|A_\gamma|, |B_\gamma| \leq K_c$ and a gap $\delta > 0$ above $\alpha \in P$, the two normalised remainders satisfy for $N \geq e$

$$\left| \frac{Z(N) - L_{<\alpha}(N)}{N^{-\alpha} \log N} - A_\alpha \right| \leq \text{errA}(N), \quad \left| \frac{Z(N) - L_{<\alpha}(N) - N^{-\alpha} A_\alpha \log N}{N^{-\alpha}} - B_\alpha \right| \leq \text{errB}(N),$$

with explicit rates $\text{errA}(N) = 2(KN^{\alpha-E} + |P|K_cN^{-\delta}) + K_c/\log N$ and $\text{errB}(N) = (KN^{\alpha-E} + |P|K_cN^{-\delta})(1 + \log N)$, both tending to 0 (`normalised_A_le`, `normalised_B_le`, `tendsto_errA`, `tendsto_errB`). Zero `sorry/axiom`.

1 The things formalised

```
def lowerPart (P : Finset ) (A B :  → ) ( N : ) :  :=
  P.filter (· < ) , N ^ (-) * (A  * log N + B )
def upperPart ... ( < ·) ...
theorem sum_split (h :  P) :
  P , ... = lowerPart ... + N^(-) * (A  * log N + B ) + upperPart ...
theorem abs_upperPart_le (hN : 1  N) (hKc) (hgap :  P,  <  →  + ) (hAB) :
  |upperPart P A B  N|  P.card * (N ^ (- ( + )) * (Kc * (1 + log N)))
def errA (K Kc E  ) (P) (N) := 2 * (K * N^(- E) + P.card * Kc * N^(-)) + Kc / log N
def errB ... := (K * N^(- E) + P.card * Kc * N^(-)) * (1 + log N)
theorem normalised_A_le (hK hKc) (h :  P) (hgap)
  (hZ :  N, 1  N → |Z N -  P, N^(-) * (A  * log N + B )|
    K * (N^(-E) * (1 + log N)))
  (hAB :  P, |A | Kc |B | Kc) (N) (hN : exp 1  N) :
  |(Z N - lowerPart P A B  N) / (N^(-) * log N) - A |  errA K Kc E  P N
theorem normalised_B_le ... :
  |(Z N - lowerPart P A B  N - N^(-) * (A  * log N)) / N^(-
) - B |  errB K Kc E  P N
theorem tendsto_rpow_neg_mul_one_add_log (a) (ha : 0 < a) :
  Tendsto (fun N => N^(-a) * (1 + log N)) atTop ( 0)
theorem tendsto_errA / tendsto_errB (hE :  < E) (h : 0 < ) : Tendsto (err...) atTop ( 0)
```

2 Mathematics

Split the finite expansion at α : $\sum_P = L_{<\alpha} + N^{-\alpha}(A_\alpha \log N + B_\alpha) + U_{>\alpha}$. The upper part is bounded by $|P| N^{-(\alpha+\delta)} K_c (1 + \log N)$ because every exponent above α is at least $\alpha + \delta$ and $N \geq 1$.

Then $(Z - L_{<\alpha})/(N^{-\alpha} \log N) - A_\alpha = (R + U)/(N^{-\alpha} \log N) + B_\alpha/\log N$, and for $\log N \geq 1$ the factor $(1 + \log N)/\log N \leq 2$ gives errA. The B-slot has no $1/\log N$ term but keeps the $(1 + \log N)$ growth, harmless against $N^{-\delta}$. The rates tend to zero by $N^{-a} \rightarrow 0$ and $\log N = o(N^a)$ (Mathlib `isLittle0_log_rpow_atTop`).

3 Paper-facing meaning

The paper's second conclusion in `thm:strataempiricalexpansion` is that the normalised remainder converges in distribution to $C_{\mu',m'}(G)$. Astra #11 stressed two points that this unit makes precise: the two slots ($\log N$ coefficient and constant) must be treated in order, and the B-slot must subtract the CURRENT random coefficient $A_\alpha(\xi_n)$, not its limit. With the uniform Taylor tree (unit 135) supplying K independent of the data on a bounded family and the envelope bounds (unit 134) supplying K_c , the deterministic error is uniform over the family, so the probabilistic statement reduces to the continuous-mapping transfer (unit 147) plus a deterministic $o(1)$ perturbation (Slutsky: `TendstoInDistribution.add_of_tendstoInMeasure_const`). That wrapper needs the measurability of $\omega \mapsto Z_n(\omega)$, i.e. joint continuity of the amplitude coefficients in (a, s) on `CoeffPair` $\times \mathbb{R}$: the next unit.

4 Lean notes

- Splitting a `Finset` sum at a member: `Finset.add_sum_erase` then `Finset.sum_filter_add_sum_filter_not` on the erased set, with two small `Finset.ext` identifications of the filters.
- `Real.add_one_le_exp 0` does not give $1 \leq e$ in a usable form; use `Real.one_le_exp zero_le_one`. $1 \leq \log N \iff e \leq N$ is `Real.le_log_iff_exp_le`.
- Powers: prove $N^{(-E)} * N^{(-)} = N^{(-E)}$ once via `Real.rpow_add` and rewrite backwards (`rw [← hE1, ← hE2]; ring`) to pass from N^{-E} to $N^{\alpha-E}N^{-\alpha}$.
- `div_le_iff` turns the normalised bound into a product bound; `field_simp`; `ring` closes the two algebraic identities for the normalised remainders.
- `Real.tendsto_log_atTop.inv_tendsto_atTop.const_mul Kc` with `simpa [div_eq_mul_inv]` gives $K_c/\log N \rightarrow 0$.